Supplementary Material for Adaptive quantum tomography by least squares with optimal asymptotic covariance

Hongru Zhao[1] **and Robert Oliver VandenBerg**[2]

[1]*School of Statistics, University of Minnesota, e-mail:* zhao1118@umn.edu

[2]*Department of Mathematics & Statistics, Bucknell University, e-mail:* rov002@bucknell.edu

This supplement is organized into two parts. Part I (Sections S1–S2) contains proofs of the main statistical results and related geometric and measurement derivations. Part II (Sections S3–S10) provides auxiliary results (Sections S3–S8), computational requirements, and reproducibility details for Figures 1 and 2 (Sections S9–S10). The pure-state geometry and conditional-likelihood framework are presented in Appendices A and B of the main article. Cross-references with an S prefix refer to this supplement; those without an S prefix refer to the main article, including its appendices.

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Part I. Proofs for the main article

S1. Proofs of the main statistical results

Each theorem and corollary of the main article has a separate proof in this section. Propositions 2.1 and 4.1 are proved in Sections S2.1 and S2.4, respectively. The conditional-likelihood framework is in Appendix B of the main article. The supporting lemmas and finite constructions are given in the remaining sections of this supplement.

S1.1. Adaptive Gill–Massar bound (Theorem 5.1)

The conditional likelihood, score convention and conditional information are given in Appendix B of the main article.

Fix the policy functions (ζ_t), and let $\mathsf{H}_{t-1} = (O_1, \ldots, O_{t-1})$ and $\mathcal{F}_{t-1} = \sigma(\mathsf{H}_{t-1})$. The past record determines $\mathsf{M}_t = \zeta_t(\mathsf{H}_{t-1})$, so M_t is $\mathcal{F}_{t-1}$-measurable. Earlier POVMs are recovered similarly and add no information to the history.

Proof of Theorem 5.1. For a fixed past record h, the policy value $\zeta_t(\mathsf{h}) = (M_{t,o}(\mathsf{h}))_o$ is an arbitrary finite single-copy POVM, with masses $\mathrm{tr}\{\rho(\theta)M_{t,o}(\mathsf{h})\}$. By Gill and Massar (2000, Theorems I and II), its conditional information satisfies $\mathrm{tr}\{H^{-1}\mathcal{I}_t(\mathsf{h})\} \le D - 1$. Section S2 also derives inequality (2.8) for both models. This holds for every h, including when the measurement is selected adaptively.

All expectations in this proof are at θ_0. For each fixed history, differentiate the finite sum of conditional masses. This gives $\mathbb{E}(s_t \mid \mathcal{F}_{t-1}) = 0$, including zero-probability outcomes: a nonnegative differentiable mass has zero derivative where it vanishes. On positive-probability records the product rule gives the joint score $S_n = \sum_{t\le n} s_t$. A record of zero probability also has zero derivative. Orthogonality of the martingale score increments yields

$$J_n = \mathbb{E}(S_n S_n^{\mathsf{T}}) = \sum_{t\le n} \mathbb{E}\{\mathcal{I}_t(\mathsf{H}_{t-1})\}, \qquad \mathrm{tr}(H^{-1}J_n) \le n(D-1). \tag{S1.1}$$

For $e = T_n - \theta_0$, finite-sum differentiation and local unbiasedness give $\partial_{\theta^{\mathsf{T}}}\mathbb{E}_\theta T_n|_{\theta_0} = \mathbb{E}(T_n S_n^{\mathsf{T}}) = \mathrm{I}_p$; since $\mathbb{E}S_n = 0$,

$$\mathbb{E}(eS_n^{\mathsf{T}}) = \mathrm{I}_p. \tag{S1.2}$$

Let $C = (H^{-1/2}WH^{-1/2})^{1/2}$. By (S1.2), $\mathbb{E}\langle CH^{1/2}e, H^{-1/2}S_n\rangle = \mathrm{tr}\,C$. Since $H^{1/2}C^2H^{1/2} = W$, Cauchy–Schwarz and (S1.1) give

$$(\mathrm{tr}\,C)^2 \le \mathbb{E}(e^{\mathsf{T}}We)\ \mathrm{tr}(H^{-1}J_n) \le n(D-1)\,\mathbb{E}(e^{\mathsf{T}}We).$$

Dividing by $D - 1$ gives (5.1), with $C_{\mathrm{GM}}(W, H)$ defined in (2.9), for every $W \succeq 0$ without inverting J_n. For pure states, $p = 2d$ and $D - 1 = d$. With $W_B = H/4$,

$$(H^{-1/2}W_BH^{-1/2})^{1/2} = \frac{1}{2}\mathrm{I}_{2d},$$

whose trace is d. Hence $C_{\mathrm{GM}}(W_B, H) = d^2/d = d$.

Similarly, with $W_F = H/2$,

$$(H^{-1/2} W_F H^{-1/2})^{1/2} = \frac{1}{\sqrt{2}} \mathrm{I}_{2d},$$

whose trace is $\sqrt{2}\, d$, giving $C_{\mathrm{GM}}(W_F, H) = 2d^2/d = 2d$.

For an interior mixed qubit, put $s_0 = (1 - \|\theta_0\|^2)^{1/2}$. The eigenvalues of $H^{-1/2}$ are $s_0, 1, 1$, so the quadratic loss weights $W_B = H/4$ and $W_F = \mathrm{I}_3/2$ give

$$C_{\mathrm{GM}}(W_B, H) = \{\mathrm{tr}(\mathrm{I}_3/2)\}^2 = 9/4,$$
$$C_{\mathrm{GM}}(W_F, H) = \{\mathrm{tr}(H^{-1/2})\}^2/2 = (2 + s_0)^2/2.$$

□

S1.2. Mixed-qubit limits (Theorems 3.1 and 3.2)

Proof of Theorem 3.1. Write

$$Q_t = \mathbb{E}(a_t a_t^{\mathsf{T}} \mid \mathcal{F}_{t-1}), \qquad M_n = \sum_{t \le n} a_t (Y_t - a_t^{\mathsf{T}} \theta_0).$$

Both M_n and $G_n - \sum_{t \le n} Q_t$ have bounded martingale increments. Under the Bures allocation, $Q_t = \mathrm{I}_3/3$ for every choice of measurement axes. Their martingale strong laws and the ridge identity

$$(\lambda \mathrm{I}_3 + G_n)(\widehat{\theta}_n - \theta_0) = M_n - \lambda \theta_0 \tag{S1.3}$$

prove consistency directly.

For the Frobenius allocation, the safeguard gives $\min_k \pi_{F,k}(c_{n_-(t)}) \ge t^{-1/4}/3$. Consequently $\sum_{t \le n} Q_t \succeq n^{3/4} \mathrm{I}_3/3$ even when a batch is arbitrarily long. Lemma S3.1 makes both centered sums $O_{\mathrm{a.s.}}(\{n \log(n+1)\}^{1/2}) = o(n^{3/4})$. Applying (S1.3) proves consistency before making any assumption about a limiting design. Since the truth is interior and $R_n \to 1$, both safeguards are eventually inactive; hence $c_n \to \theta_0$.

Positive finite batches give $n_-(t) \to \infty$, so the estimate used to select the next axes also tends to θ_0. At its own design estimate the conditional information equals the continuous formula $\mathcal{I}_L^\star(c_{n_-(t)})$. On each smaller Bloch ball, $\mathcal{I}_a(\theta)$ is uniformly Lipschitz in θ over all unit axes. Replacing the design estimate by the truth therefore changes this information by $o(1)$. This proves the current-information limit, including at zero truth, without convergence of the chosen orthonormal axes. Cesàro averaging gives the conditional average; the realized-setting average follows because $\mathcal{I}_{a_t}(\theta_0) - \mathcal{I}_{\xi_t}(\theta_0)$ is a bounded martingale difference.

The strong laws and averaging used here are proved in Lemmas S3.1 and S3.2. □

Proof of Theorem 3.2. Use Q_t and M_n from the preceding proof. Let V_t be the conditional covariance of the increment of M_n. For nonzero truth put $\Pi_r = \theta_0 \theta_0^{\mathsf{T}} / \|\theta_0\|^2$ and $\Pi_\perp = \mathrm{I}_3 - \Pi_r$. If $(p_r, p_\perp)$ are the radial and each transverse allocation in Section 3.1, then

$$Q_t \longrightarrow Q_L^\star = p_r \Pi_r + p_\perp \Pi_\perp,$$
$$V_t \longrightarrow V_L^\star = p_r s_0^2 \Pi_r + p_\perp \Pi_\perp.$$

The same replacement argument proves these limits; the two transverse weights are equal, so the transverse basis is immaterial. Lemma S3.1 and averaging give $G_n/n \to Q_L^\star$. Lemma S3.3

gives $M_n/\sqrt{n} \Longrightarrow \mathcal{N}_3(0, V_L^\star)$. The ridge identity (S1.3) and Slutsky's theorem therefore give covariance

$$(Q_L^\star)^{-1} V_L^\star (Q_L^\star)^{-1} = \frac{s_0^2}{p_r}\Pi_r + \frac{1}{p_\perp}\Pi_\perp = (\mathcal{I}_L^\star)^{-1}. \tag{S1.4}$$

At zero truth, $s(c_{n_(t)}) \to 1$ makes both allocations approach $(1/3, 1/3, 1/3)$; hence $Q_L^\star = V_L^\star = \mathrm{I}_3/3$ for every choice of axes, giving $3\mathrm{I}_3$. The fixed ridge term vanishes at the $\sqrt{n}$ scale. Eventual equality of c_n and $\widehat{\theta}_n$ transfers asymptotic normality to the physical output. A deterministic subsequence gives the completed-batch limits. The bounded-increment central limit argument is Lemma S3.3. □

S1.3. Pure-state limits (Theorems 4.2 and 4.3)

Proof of Theorem 4.2. Use the frame operators and ridge normal equations (S2.2) and (S2.3) from Section S2.4. The conditional Born law makes $E_{t,Z_t} - \mathcal{S}_{U_t}(P_0)$ a bounded Hermitian martingale difference. The preliminary ridge equation and (C1) give

$$G_n(\widehat{A}_n - P_0) = \sum_{t \le n} \{E_{t,Z_t} - \mathcal{S}_{U_t}(P_0)\} - \lambda P_0, \qquad G_n \succeq (\lambda + n\lambda_{\min}(S_{\mathsf{E}}))\mathrm{I}.$$

The martingale strong law proves consistency of $\widehat{A}_n$. Nearest-pure projection increases the error by at most a factor two, so $\widehat{P}_n \to P_0$ almost surely as well.

For information convergence, every subsequential limit of the selected unitaries aligns P_{ref} with P_0, by consistency at batch starts and compactness of the unitary group. Any two such aligned unitaries differ by a phase and a unitary on the orthogonal complement. Their real action on the normalized tangent coordinates is orthogonal, so it preserves $\mathcal{I} = \mathrm{I}_{2d}/2$ from Proposition 4.1. Condition (C2) makes the finite Fisher sum continuous at each aligned limit and gives eventual positive probabilities. Every subsequential information limit is therefore $\mathcal{H}_{P_0}/2$; the full unitary need not converge. Cesàro averaging gives the accumulated limit. At every aligned limit the outcome probabilities are $\mathrm{tr}(P_{\mathrm{ref}} E_z) \ge 4\tau$. Compactness therefore makes all true outcome probabilities at least 2τ eventually almost surely.

To prove strong consistency of the corrected output, let $e_n = \widehat{P}_n - P_0$ and let M_n^0 be the residual sum with inverse truncated probabilities evaluated at P_0. Its increments are bounded martingale differences, so $M_n^0/n \to 0$ almost surely. The function $w(p) = 1/\max(p, \tau)$ is bounded and Lipschitz; hence the fitted score differs from this true-state score by at most $Cn\|e_n\|_F$. The fitted tangent normal operator is bounded below by $(\eta + n\lambda_{\min}(S_{\mathsf{E}}))\mathrm{I}$. Consequently

$$\left\|\widehat{\Delta}_n\right\|_F \le C\{\left\|M_n^0\right\|_F/n + \|e_n\|_F\} \longrightarrow 0 \quad \text{almost surely.}$$

The global retraction bound in Lemma S4.1 then gives $\widehat{P}_n^{\mathrm{eff}} \to P_0$ almost surely. Completed batches form a deterministic subsequence. The probability tools are Lemmas S3.1 and S3.2. □

Proof of Theorem 4.3. Use the residual martingale and uniform Gram bound from the preceding proof. Martingale orthogonality gives a second moment of order n, so the ridge equation yields $\widehat{A}_n - P_0 = O_p(n^{-1/2})$. The nearest-pure projection bound then gives $\widehat{P}_n - P_0 = O_p(n^{-1/2})$.

This proves part (i). Theorem 4.2 supplies consistency, the information limit and the eventual probability bound required by Lemma S6.2; we now apply it to prove part (ii).

Put $e_n = \widehat{P}_n - P_0$, $e_{T,n} = \Pi_{\widehat{P}_n} e_n$, and let A_n be the fitted tangent normal operator defined in Section S6, whose coordinate matrix is $\mathcal{K}_n$ in (S2.4). For the complete outcome label z, define the ambient score evaluated at the truth by

$$S_n^0 = \sum_{t \le n, z} \frac{E_{t,z}\{\mathbf{1}_{\{Z_t = z\}} - p_{t,z}\}}{\max(p_{t,z}, \tau)}, \qquad p_{t,z} = \operatorname{tr}(E_{t,z} P_0).$$

This is a bounded-increment martingale. Lemma S6.2 applies by part (i) and Theorem 4.2, giving

$$\widehat{\Delta}_n = -e_{T,n} + \mathsf{A}_n^{-1} \Pi_{\widehat{P}_n} S_n^0 + O_p(n^{-1}). \tag{S1.5}$$

The same lemma gives $n\mathsf{A}_n^{-1}\Pi_{\widehat{P}_n} \to \Pi_{P_0}$ almost surely. Its pathwise bounds account for the finite initial clipped records without conditioning on the preliminary fit or treating its fitted weights as predictable.

Work in fixed true coordinates whose tangent basis is Frobenius-orthonormal. After the finite initial clipped set, all probabilities exceed the floor. Let $g_{t,z}$ be the coordinate gradient of $\operatorname{tr}\{E_{t,z}\Gamma(\theta)\}$ at $\theta = \theta_0$, and put $\varepsilon_{t,z} = \mathbf{1}_{\{Z_t = z\}} - p_{t,z}$. Applying Lemma S6.1 to the true probabilities gives

$$\begin{aligned} \operatorname{Cov}\left(\sum_z \frac{g_{t,z}}{p_{t,z}} \varepsilon_{t,z} \,\middle|\, \mathcal{F}_{t-1} \right) &= \sum_z \frac{g_{t,z} g_{t,z}^{\mathsf{T}}}{p_{t,z}} \\ &= \mathcal{I}_{\xi_t}(\theta_0). \end{aligned} \tag{S1.6}$$

Information convergence makes this covariance tend to the tangent identity. The floor bounds all score increments, so Lemma S3.3 yields $n^{-1/2}\Pi_{P_0} S_n^0 \Longrightarrow \mathcal{N}(0, \mathrm{I}_{T_{P_0}})$. The fitted normalized normal operator tends to the tangent identity. Since the normalization retraction has a quadratic error,

$$\widehat{P}_n^{\mathrm{eff}} - P_0 = e_{T,n} + \widehat{\Delta}_n + O_p(n^{-1}) = n^{-1}\Pi_{P_0} S_n^0 + o_p(n^{-1/2}).$$

This cancels the preliminary error and proves asymptotic normality. In normalized amplitude coordinates the asymptotic covariance is $2\mathrm{I}_{2d}$; Section S2.1 and Lemma S4.2 give $2H^{-1}$ in any smooth chart.

Completed-batch limits follow by a deterministic subsequence. □

The next corollary uses asymptotic normality of the pure-state estimator to derive limiting distributions for infidelity and squared Frobenius loss.

Corollary S1.1 (Pure-state loss distributions)**.** *Under the hypotheses of Theorem 4.3, let* $\overline{P}_n = \widehat{P}_n^{\mathrm{eff}}$ *be the phase-paired output in* (4.9)*. Then*

$$\begin{aligned} n[1 - F(\overline{P}_n, P_0)] &\Longrightarrow \tfrac{1}{2}\chi^2_{2d}, \\ n\left\|\overline{P}_n - P_0\right\|_F^2 &\Longrightarrow \chi^2_{2d}. \end{aligned} \tag{S1.7}$$

Proof of Corollary S1.1. In the normalized true amplitude chart the two losses are exactly $\|\theta\|^2/4$ and $\|\theta\|^2/2$, respectively. Theorem 4.3 and the continuous mapping theorem give the result. Consistency makes this chart valid eventually; definitions outside it do not affect the limit. This argument is distributional and does not interchange expectations and limits. □

S1.4. Expected-loss attainment (Corollaries 5.2 and 5.3)

Propositions S7.2 and S7.3, with $r = 4$, prove the common bound used for Corollaries 5.2 and 5.3:

$$\sup_{n\geq 1} n^2 \mathbb{E}_{\rho_0} \|\widetilde{\rho}_n - \rho_0\|_F^4 < \infty, \tag{S1.8}$$

where $\widetilde{\rho}_n$ is the relevant physical output. Constants may depend on the fixed truth, dimension and parameters; no uniformity toward the mixed-state boundary is claimed. For the pure preliminary fits, Proposition S7.3 also gives, for every fixed $r \geq 2$,

$$\mathbb{E}\left\|\widehat{A}_n - P_0\right\|_F^r + \mathbb{E}\left\|\widehat{P}_n - P_0\right\|_F^r \leq C_r n^{-r/2}. \tag{S1.9}$$

The proposition also proves (S1.8) for the corrected output, including its truncated weights, tangent solve and normalization.

Proof of Corollary 5.2. Proposition S7.2 establishes (S1.8) for both physical outputs. Theorem 3.2 and the local quadratic loss expansion give

$$nL(\widetilde{\rho}_n, \rho_0) \Longrightarrow Z^{\mathsf{T}} W_L Z, \qquad Z \sim \mathcal{N}_p(0, \Sigma_L), \qquad \Sigma_L = (\mathcal{I}_L^{\star})^{-1}. \tag{S1.10}$$

All three matrices use the same local coordinates. For Frobenius loss, $L = \|\widetilde{\rho}_n - \rho_0\|_F^2$ exactly. For mixed qubits there is also a direct identity, valid for every physical estimate, including a pure estimate. Write u for its Bloch vector, $\delta = u - \theta_0$, $s(u) = \{1 - \|u\|^2\}^{1/2}$ and $s_0 = s(\theta_0) > 0$. Rationalizing the exact fidelity formula gives

$$D_B^2 = \frac{\|\delta\|^2 + \{(u+\theta_0)^{\mathsf{T}}\delta/[s_0 + s(u)]\}^2}{2(1+\sqrt{F})}.$$

The coefficients are continuous in u, and at $u = \theta_0$ the right-hand side is the quadratic form $\delta^{\mathsf{T}} H(\theta_0)\delta/4$. Moreover, $\|u + \theta_0\| \leq 2$ and $s_0 + s(u) \geq s_0$, so $D_B^2 \leq (1 + 4/s_0^2)\|\rho(u) - \rho(\theta_0)\|_F^2$ globally. Thus (S1.8) bounds the second moments of nL and makes these losses uniformly integrable. More explicitly, write $X_n = nL$ and $X = Z^{\mathsf{T}} W_L Z$. For each fixed $R > 0$, weak convergence gives $\mathbb{E}\min(X_n, R) \to \mathbb{E}\min(X, R)$, since the function $x \mapsto \min\{\max(x, 0), R\}$ is bounded and continuous. Moreover,

$$0 \leq \mathbb{E}X_n - \mathbb{E}\min(X_n, R) \leq \frac{\mathbb{E}X_n^2}{R}, \qquad 0 \leq \mathbb{E}X - \mathbb{E}\min(X, R) \leq \frac{\mathbb{E}X^2}{R}.$$

The first numerator is bounded uniformly in n and the second is finite because X is a Gaussian quadratic form. First let $n \to \infty$ and then $R \to \infty$. This justifies taking expectations in (S1.10) and yields

$$n\mathbb{E}_{\rho_0} L(\widetilde{\rho}_n, \rho_0) \longrightarrow \operatorname{tr}(W_L \Sigma_L) = C_{\mathrm{GM}}(W_L, H).$$

The two matched allocations give $9/4$ and $(2 + s_0)^2/2$, as asserted. □

Proof of Corollary 5.3. Proposition S7.3 establishes (S1.8) for $\widehat{P}_n^{\mathrm{eff}}$. Theorem 4.3 and the local loss expansions give the Gaussian quadratic limit (S1.10) with $\Sigma_L = 2H^{-1}$, $W_B = H/4$ and $W_F = H/2$. For pure truth and pure output, $D_B^2 \leq \|\widehat{P}_n^{\mathrm{eff}} - P_0\|_F^2$; hence (S1.8) bounds the second moments of both scaled losses. The truncation argument in the preceding proof therefore

justifies convergence of expectations. Since the chart dimension is $2d$, the limits $\operatorname{tr}(W_B\Sigma_L) = d$ and $\operatorname{tr}(W_F\Sigma_L) = 2d$ follow. The finite-sample outputs may be biased: local unbiasedness defines the oracle comparison class in Theorem 5.1, not an additional requirement on these outputs. □

Proof of the Table 1 attainment claims. For mixed-qubit pairs and pure-state pairs, the difference has rank at most two and trace zero. Its two possible nonzero eigenvalues are opposite, so $\|\rho - \sigma\|_1^2 = 2\|\rho - \sigma\|_F^2$ exactly. The squared full trace-norm limits therefore follow from the Frobenius limits. For infidelity, the conventions (2.7) give

$$D_B^2 - (1 - F) = (1 - \sqrt{F})^2 = \tfrac{1}{4}(D_B^2)^2.$$

The global Bures bounds in the preceding corollary proofs and (S1.8) imply $n\mathbb{E}\{D_B^2 - (1 - F)\} = O(n^{-1}) \to 0$. Thus the infidelity limits equal the corresponding Bures limits. □

S2. Geometric and measurement arguments

S2.1. Pure-state quantum Fisher information (Proposition 2.1)

Proof of Proposition 2.1. At $P = \Gamma(\theta)$, a tangent vector $A \in T_P$ is the derivative $A = \gamma'(0)$ of a smooth curve $\gamma(s) \in \mathcal{P}_D$ with $\gamma(0) = P$. Differentiating $\gamma(s)^2 = \gamma(s)$ gives $PA + AP = A$. In a chart, write $\gamma(s) = \Gamma(\theta(s))$ with $\theta(0) = \theta$ and $A_j = \partial_j\Gamma(\theta)$. Then $A = \sum_{j=1}^p u_j A_j$, where $u_j = \theta_j'(0) \in \mathbb{R}$. For chosen coordinate SLDs, the directional SLD $\Lambda_A = \sum_{j=1}^p u_j\Lambda_j$ satisfies

$$A = \frac{P\Lambda_A + \Lambda_A P}{2}.$$

The choice $\Lambda_j = 2A_j$ gives $\Lambda_A = 2A$, which solves this equation because $PA+AP = A$ (Facchi et al., 2010, Eqs. (17)–(19)). Similarly, for $B = \sum_{k=1}^p w_k A_k$ with $w \in \mathbb{R}^p$, $\Lambda_B = \sum_{k=1}^p w_k\Lambda_k$ gives the canonical choice $\Lambda_B = 2B$.

If another directional SLD is $2A + K_A$, its Hermitian difference satisfies $PK_A + K_AP = 0$. Multiplication by P on both sides gives $PK_AP = 0$, and hence $PK_A = K_AP = 0$. The same argument applies to K_B. Cyclicity of the trace then shows that these differences do not change the pairing. With the canonical SLDs,

$$\begin{aligned}\mathcal{H}_P[A, B] &= \frac{1}{2}\operatorname{tr}\{P(\Lambda_A\Lambda_B + \Lambda_B\Lambda_A)\} \\ &= 2\operatorname{tr}\{P(AB + BA)\} = 2\operatorname{tr}(AB).\end{aligned}$$

The last equality follows by multiplying $PA + AP = A$ by B and taking the trace. At $P = hh^*$ with $\|h\| = 1$ and $h^*v = 0$, the tangent vector $A = hv^* + vh^*$ satisfies $\operatorname{tr}(A^2) = 2\|v\|^2$, so $\mathcal{H}_P[A, A] = 4\|v\|^2$. This proves (2.5).

Finally, applying the pairing to A_j, A_k gives $[H(\theta)]_{jk} = \mathcal{H}_P[A_j, A_k]$ directly from the coordinate SLD definition. Bilinearity yields $\mathcal{H}_P[A, B] = u^{\mathsf{T}}H(\theta)w$ for the coordinate representations of A and B above. □

S2.2. Qubit loss identities

For a Bloch difference δ, the eigenvalues of the density-matrix difference are $\pm\|\delta\|/2$. Hence

$$\|\rho-\sigma\|_1^2 = \|\delta\|^2 = 2\,\|\rho-\sigma\|_F^2.$$

The usual squared trace distance is one quarter of the first quantity. For an interior truth and $s_0 = \{1-\|\theta_0\|^2\}^{1/2}$, the exact qubit fidelity formula (Miszczak et al., 2009) is

$$F = \tfrac{1}{2}\left[1+\theta_0^{\mathsf{T}}(\theta_0+\delta)+s_0\{1-\|\theta_0+\delta\|^2\}^{1/2}\right].$$

Taylor expansion gives $1-F = \delta^{\mathsf{T}}H(\theta_0)\delta/4 + O(\|\delta\|^3)$. Also $D_B^2-(1-F) = (1-F^{1/2})^2 = O(\|\delta\|^4)$. These statements give the loss weights in Section 2.5; they do not identify all finite-separation Bures and infidelity optimization problems.

S2.3. Single-copy information bounds

For completeness, recall the elementary derivation of the weighted Gill–Massar bound used in Section 2.5 (Zhu and Hayashi, 2018, Supplemental Material, Section II B). For positive definite $H, \mathcal{I}$ and positive semidefinite W, set $A = H^{-1/2}WH^{-1/2}$ and $B = H^{-1/2}\mathcal{I}H^{-1/2}$. Cyclicity of trace gives $\operatorname{tr}(W\mathcal{I}^{-1}) = \operatorname{tr}(AB^{-1})$. Hilbert–Schmidt Cauchy–Schwarz, applied to $B^{-1/2}A^{1/2}$ and $B^{1/2}$, gives

$$(\operatorname{tr}A^{1/2})^2 \le \operatorname{tr}(AB^{-1})\operatorname{tr}B \le (D-1)\operatorname{tr}(W\mathcal{I}^{-1}).$$

This proves (2.9) whenever (2.8) holds. For $W \succ 0$, equality in the positive-definite matrix relaxation occurs at $B = (D-1)A^{1/2}/\operatorname{tr}A^{1/2}$: it has trace $D-1$ and makes the two matrices in Cauchy–Schwarz proportional. For singular nonzero W, this expression is singular; for $W=0$, the inequality is trivial. This argument gives the information bound; attaining it requires a measurement realizing the resulting information and an estimator attaining its inverse. We next verify the trace constraint and the loss-specific equality cases used in the paper.

Every nonzero qubit effect can be written $E_y = \alpha_y(\mathrm{I}_2 + v_y\cdot\sigma)$, where $\alpha_y > 0$, $\|v_y\| \le 1$, $\sum_y \alpha_y = 1$, and $\sum_y \alpha_y v_y = 0$. With $H^{-1} = \mathrm{I}_3 - \theta_0\theta_0^{\mathsf{T}}$,

$$\begin{aligned}\operatorname{tr}(H^{-1}\mathcal{I}) &= \sum_y \alpha_y \frac{\|v_y\|^2-(v_y^{\mathsf{T}}\theta_0)^2}{1+v_y^{\mathsf{T}}\theta_0}\\ &\le \sum_y \alpha_y(1-v_y^{\mathsf{T}}\theta_0) = 1.\end{aligned}$$

This is the qubit Gill–Massar tradeoff (Gill and Massar, 2000). The relative information $H^{-1/2}\mathcal{I}H^{-1/2}$ is positive semidefinite with trace at most one. Each of its eigenvalues therefore lies in $[0,1]$, equivalently $0 \preceq \mathcal{I} \preceq H$. For Bures loss, the eigenvalues of $H^{-1/2}\mathcal{I}H^{-1/2}$ sum to at most one, so $\operatorname{tr}[(H/4)\mathcal{I}^{-1}] \ge 9/4$, with equality at $\mathcal{I} = H/3$. For Frobenius loss, Hilbert–Schmidt Cauchy–Schwarz gives

$$\begin{aligned}(\operatorname{tr}H^{-1/2})^2 &\le \operatorname{tr}(\mathcal{I}^{-1})\operatorname{tr}(H^{-1}\mathcal{I})\\ &\le \operatorname{tr}(\mathcal{I}^{-1}).\end{aligned}$$

The eigenvalues of $H^{-1/2}$ are $s_0, 1, 1$. Thus the minimum Frobenius criterion is $(2+s_0)^2/2$, attained by (3.5). This reproduces the known qubit allocations while establishing the comparison class used in Theorem 3.1.

For a pure-state effect at the reference state write $E_y = \begin{pmatrix} p_y & b_y^* \\ b_y & C_y \end{pmatrix}$. In the normalized chart, the probability gradient is $(\Re b_y, \Im b_y)$. If $p_y > 0$, positivity implies $b_y b_y^* \preceq p_y C_y$. Consequently

$$\operatorname{tr}(H^{-1}\mathcal{I}) = \sum_{y:p_y>0} \|b_y\|^2/p_y \le \sum_y \operatorname{tr} C_y = d. \tag{S2.1}$$

If $p_y = 0$, positivity forces $b_y = 0$ and the outcome contributes zero under the positive-support convention in Section 2.4. For any normalized tangent direction, the same inequality bounds its scalar Fisher contribution by the corresponding quadratic form of C_y. Completeness then gives $0 \preceq \mathcal{I} \preceq H$. In particular, early zero outcome probabilities do not make the finite-outcome Fisher matrix diverge.

In coordinates normalized by the quantum Fisher information, nonsingular information satisfies

$$\operatorname{tr}(\mathcal{I}^{-1}) \ge \frac{(2d)^2}{\operatorname{tr}\mathcal{I}} \ge 4d.$$

Equality holds at $\mathcal{I} = \mathrm{I}_{2d}/2$. The local infidelity weight is $\mathrm{I}_{2d}/4$, giving d; pure-state Frobenius loss gives $2d$. For a measurement with positive reference probabilities, equality of the trace budget in (S2.1) makes every positive semidefinite Schur complement $C_y - b_y b_y^*/p_y$ vanish. Each nonzero effect is then rank one. This explains the rank-one hypothesis in the phase-pair construction; noisy effects of higher rank can still identify the state without achieving this information budget.

S2.4. Phase pairing and uniform least-squares bounds

Proof of Proposition 4.1. Completeness follows by adding the two halves; the first half already spans the Hermitian space. At P_{ref}, both branch probabilities are $p_r^{\mathrm{ref}}/2 > 0$ by (C2), since $VP_{\mathrm{ref}}V^* = P_{\mathrm{ref}}$. Write $F_r = w_r w_r^*$ and choose its phase so that $w_r = \{p_r^{\mathrm{ref}}\}^{1/2} e_0 + v_r$, with $v_r \perp e_0$. The complement block of completeness is

$$\sum_r v_r v_r^* = \mathrm{I}_d.$$

With $g(v) = (\Re v, \Im v)$, differentiation of the Born law in (4.1) gives gradient $\{p_r^{\mathrm{ref}}\}^{1/2} g(v_r)$ for the reference measurement outcome. For the state gate $V = \mathrm{I}_D + (i-1)P_{\mathrm{ref}}$ in the main article,

$$V^* w_r = -i\{p_r^{\mathrm{ref}}\}^{1/2} e_0 + v_r = -i\big(\{p_r^{\mathrm{ref}}\}^{1/2} e_0 + i v_r\big).$$

The scalar $-i$ cancels in the rank-one effect. Hence the phase partner again replaces v_r by iv_r. The block identity

$$g(v)g(v)^{\mathsf{T}} + g(iv)g(iv)^{\mathsf{T}} = \begin{pmatrix} \Re(vv^*) & -\Im(vv^*) \\ \Im(vv^*) & \Re(vv^*) \end{pmatrix}$$

and the recorded fair choice prove (4.4). □

Proof of Remark 1. Take the informationally complete reference family from Proposition S5.1 and one of its component PVMs B. A recorded mixture with probabilities ϵ and $1-\epsilon$, respectively, satisfies (C1)–(C2) for every $0 < \epsilon < 1$. Its reference information is $\epsilon\mathcal{I}_{\mathsf{F}}+(1-\epsilon)\mathcal{I}_{\mathsf{B}}$. The D PVM probabilities sum to one, so $\operatorname{rank}(\mathcal{I}_{\mathsf{B}}) \le D-1 < 2d$. As $\epsilon \downarrow 0$, the mixture's information tends to this singular matrix, whereas $H/2$ is nonsingular. Hence sufficiently small positive ϵ gives a seed satisfying (C1)–(C2) without Fisher symmetry. A seed already having $\mathcal{I} = H/2$ needs no balancing. □

Normal equations and uniform invertibility. For a finite POVM $\mathsf{M} = \{M_z\}$, its measurement frame operator acts on the real Hilbert space $\mathrm{Herm}(\mathbb{C}^D)$ with the Frobenius inner product. For a unitary U, define

$$\begin{aligned} S_{\mathsf{M}}(A) &= \sum_z \operatorname{tr}(M_z A) M_z, \\ \mathcal{S}_U(A) &= \sum_{z\in\mathcal{Z}} \operatorname{tr}(UE_zU^*A)UE_zU^*. \end{aligned} \tag{S2.2}$$

Taking $\mathsf{M} = \mathsf{F}$ or E gives S_{F} or S_{E}, respectively. The frame operator differs from Fisher information: it does not weight the outcome contributions by inverse probabilities.

For a finite POVM M, the effects span $\mathrm{Herm}(\mathbb{C}^D)$ if and only if $S_{\mathsf{M}} \succ 0$, with S_{M} defined in (S2.2). If F satisfies (C1), its phase pair E in (4.3) satisfies

$$S_{\mathsf{E}} = \tfrac{1}{4}\big(S_{\mathsf{F}} + \mathrm{Ad}_{V^*} \circ S_{\mathsf{F}} \circ \mathrm{Ad}_V\big) \succ 0, \qquad \mathrm{Ad}_U(A) = UAU^*.$$

For any informationally complete reference POVM E, every unitary U satisfies

$$\mathcal{S}_U = \mathrm{Ad}_U \circ S_{\mathsf{E}} \circ \mathrm{Ad}_{U^*} \succeq \lambda_{\min}(S_{\mathsf{E}})\mathrm{I}_{\mathrm{Herm}(D)}.$$

Here $\lambda_{\min}(S_{\mathsf{E}}) > 0$ is the smallest eigenvalue of S_{E}, and $\mathrm{I}_{\mathrm{Herm}(D)}$ is the identity operator $A \mapsto A$ on $\mathrm{Herm}(\mathbb{C}^D)$. The operators $\mathcal{S}_U$ and S_{E} have the same spectrum, and every rotated measurement $\{UE_zU^*\}$ is informationally complete on the full state space.

Proof of the frame-coercivity bounds. For Hermitian A, B, symmetry follows from

$$\langle A, S_{\mathsf{M}}(B)\rangle_F = \sum_z \operatorname{tr}(M_z A)\operatorname{tr}(M_z B).$$

In particular,

$$\langle A, S_{\mathsf{M}}(A)\rangle_F = \sum_z \operatorname{tr}(M_z A)^2 \ge 0.$$

Its zero set is the orthogonal complement of $\operatorname{span}_{\mathbb{R}}\{M_z\}$. Thus the effects span the Hermitian space exactly when S_{M} is positive definite; finite dimensionality then gives a positive smallest eigenvalue (Scott, 2006, Sections III–IV). Substitution of (4.3) gives the displayed identity for S_{E}. Unitary conjugation is orthogonal for the Frobenius inner product, so $\lambda_{\min}(S_{\mathsf{E}}) \ge \frac{1}{2}\lambda_{\min}(S_{\mathsf{F}}) > 0$ under (C1). Cyclicity of trace also gives $\mathcal{S}_U(A) = US_{\mathsf{E}}(U^*AU)U^*$. Orthogonal conjugation preserves the spectrum, yielding the stated uniform lower bound and, by the first assertion, identification. □

Differentiating the ridge objective (4.5) gives the normal equations

$$G_n = \lambda \mathrm{I}_{\mathrm{Herm}(D)} + \sum_{t \le n} \mathcal{S}_{U_t}, \qquad B_n = \sum_{t \le n} E_{t,Z_t}, \qquad \widehat{A}_n = G_n^{-1} B_n. \tag{S2.3}$$

Thus $G_n \succeq (\lambda + n\lambda_{\min}(S_{\mathsf{E}}))\mathrm{I}_{\mathrm{Herm}(D)}$. The ridge term alone gives finite-sample uniqueness; the uniform frame bound controls the consistency and preliminary rate in Section S1.3.

Our uncorrected baseline is this rank-one projection of the unweighted categorical least-squares estimator. Splitting an effect into equal parts with distinct labels preserves its Fisher information but reduces its contribution to the unweighted frame operator. Unequal refinements therefore change this baseline without adding information; its variance inflation is not a universal property of least-squares tomography.

S2.5. Measurement construction, alignment and projection

Proof of Example 4.1. Each basis resolves the identity, so $\sum_{\ell,k} F_{\ell,k} = \mathrm{I}_D$. Uniformly selecting ℓ and recording its readout k gives, for every input state ρ,

$$\mathbb{P}\{\ell_t = \ell, k_t = k\} = B^{-1} \operatorname{tr}(\rho f_{\ell,k} f_{\ell,k}^*) = \operatorname{tr}(\rho F_{\ell,k}).$$

The assumed spanning and nonzero overlaps give (C1)–(C2).

Necessary number of bases. Each basis resolves the identity, $\sum_{k=1}^{D} f_{\ell,k} f_{\ell,k}^* = \mathrm{I}_D$. The first basis contributes at most D independent Hermitian directions; each additional basis contributes at most $D-1$ new directions because the identity direction is shared. Hence

$$\dim_{\mathbb{R}} \operatorname{span}_{\mathbb{R}}\{f_{\ell,k} f_{\ell,k}^* : \ell \in [B],\ k \in [D]\} \le D + (B-1)(D-1) = 1 + B(D-1).$$

Since $\dim_{\mathbb{R}} \mathrm{Herm}(\mathbb{C}^D) = D^2$, condition (C1) necessarily requires

$$1 + B(D-1) \ge D^2, \qquad \text{so} \qquad B \ge D+1.$$

This dimension-counting bound is only necessary: even when $B = D+1$, the chosen projectors must actually span $\mathrm{Herm}(\mathbb{C}^D)$. In particular, $B = 1$ cannot satisfy (C1) for $D \ge 2$.

The randomized-PVM representation in Section 4 records both the PVM index and its outcome. Applying the phase gate before measurement produces $V^* F_r V$, so the fair recorded branch choice implements exactly the two effects in (4.3). Proposition S5.1 supplies an explicit collection of reference bases with the required identifying and positivity properties. □

Proof of (4.6). For completeness, the deterministic alignment in (4.6) has no vanishing denominator. In its notation, $|\gamma| = 1$ and $\overline{\gamma} e_0^* a = |e_0^* a|$, so $v^* v = 2(1 + |e_0^* a|) \in [2,4]$ and $v^* e_0 = (v^* v)/2$. Hence $U(a) e_0 = a$. The matrix $Q = v v^*/(v^* v)$ is an orthogonal projector, so $(2Q - \mathrm{I})^2 = \mathrm{I}$ and $U(a)^* U(a) = \mathrm{I}$. At batch j the input is $a_t = \widehat{h}_{N_{j-1}}$ throughout the batch. The reflector is a standard numerical construction (LAPACK, 2026); the formula fixes its phase and sign. □

Proof of (3.4). For $u \neq e_3$, put $v = u - e_3$. Then $v^{\mathsf{T}}v = 2(1 - u_3) > 0$ and $v^{\mathsf{T}}e_3 = -v^{\mathsf{T}}v/2$. Thus $Q = \mathrm{I}_3 - 2vv^{\mathsf{T}}/(v^{\mathsf{T}}v)$ is orthogonal and $Qe_3 = e_3 + v = u$. Its other two columns give the required orthonormal transverse pair. The cases $u = e_3$ and $c_n = 0$ follow from the stated reference choices. □

For Hermitian A, $\|A - hh^*\|_F^2 = \|A\|_F^2 + 1 - 2h^*Ah$ for every unit h. Thus a largest-algebraic-eigenvalue vector gives a nearest pure projector $\mathrm{proj}_{\mathcal{P}_D}(A)$, even if A is not positive semidefinite. The minimizing property gives $\|\mathrm{proj}_{\mathcal{P}_D}(A) - P_0\|_F \leq 2\|A - P_0\|_F$. The tangent and retraction bounds used in Section S1.3 are proved in Section S4; the local geometric definitions are in Appendix A.

S2.6. The preliminary-estimator covariance example

Proof of (4.7). Use the finite measurement of Section S8, with $\lambda = 1$, the preliminary estimator (4.5), and the same prescribed finite batches and alignment rule as in Section 4. Proposition S8.1 supplies ambient identification, positive reference probabilities and reference information $\mathcal{H}/2$. The frame bound in Section S2.4 applies to this informationally complete family. Its ridge equation and the bounded residual martingale give $\widehat{A}_n \to P_0$ almost surely and $\widehat{A}_n - P_0 = O_p(n^{-1/2})$. Nearest-pure projection gives the same consistency and rate for $\widehat{P}_n$. The compactness and transport argument in the proof of Theorem 4.2, now using Proposition S8.1, proves alignment and information convergence to $\mathcal{H}_{P_0}/2$.

Let $T = T_{P_0}$ and $M_n = \sum_{t \leq n}\{E_{t,Z_t} - \mathcal{S}_{U_t}(P_0)\}$. Lemma S8.2 gives, at every correctly aligned frame, tangent Gram coefficient $\alpha_D = (4D - 5)/(32Ld)$ and tangent noise variance $\beta_D = (5D - 7)/(256L^2d)$, where $L = d(4d + 1)$. These finite polynomial expressions are continuous in the selected unitary. Alignment and compactness therefore give

$$\Pi_T(G_n/n) \to \alpha_D\Pi_T, \qquad \Pi_T M_n/\sqrt{n} \Longrightarrow \mathcal{N}_T(0, \beta_D\mathrm{I}_T).$$

The first limit follows by averaging; the second uses Lemma S3.3. Projecting the exact ridge equation and using the root-n bound to control the vanishing operator error gives

$$\sqrt{n}\,\Pi_T(\widehat{A}_n - P_0) = \alpha_D^{-1}\Pi_T M_n/\sqrt{n} + o_p(1).$$

Lemma S4.1 further gives $\widehat{P}_n - P_0 = \Pi_T(\widehat{A}_n - P_0) + O_p(n^{-1})$. Thus its tangent variance is

$$\beta_D/\alpha_D^2 = \frac{4(D - 1)(5D - 7)}{(4D - 5)^2} = v_D,$$

which proves (4.7).

For the corrected estimate, fix $\eta > 0$ and set $\tau = \frac{1}{4}\min_z \mathrm{tr}(P_{\mathrm{ref}}E_z)$. The consistency, rate and information limits established above verify the assumptions of Lemma S6.2 for this family. Applying it with Lemma S6.1 and the retraction bound gives corrected tangent variance one by the same score expansion as in Section S1.3. □

S2.7. Computing the tangent correction

The quadratic objective (4.8) is solved in coordinates on the fitted tangent space. Algorithm S1 gives the normal equations and reconstructs the reported pure-state estimate. Its input is the same preliminary fit and complete setting–outcome records used in Section 4.5.

Algorithm S1 Computing the weighted tangent correction

Input. The complete records (Z_t, a_t), $t \le n$, with $Z_t \in \mathcal{Z}$ and recorded effects $E_{t,z} = E_{a_t,z}$; the preliminary unit vector $h = \widehat{h}_n$ and projector $P = \widehat{P}_n = hh^*$; the ridge parameter $\eta > 0$ and denominator floor τ from Section 4.5.

Output. The tangent correction $\widehat{\Delta}_n$ and reported estimate $\widehat{P}_n^{\mathrm{eff}}$.

1. **Choose tangent coordinates.** Choose a Frobenius-orthonormal basis $T_1, \dots, T_{2d}$ of T_P. For each $t \le n$ and $z \in \mathcal{Z}$, form $g_{t,z} \in \mathbb{R}^{2d}$ with $[g_{t,z}]_j = \mathrm{tr}(E_{t,z}T_j)$. This is the coordinate vector of $\Pi_P E_{t,z}$ and gives the outcome-probability sensitivity along each T_j.
2. **Fit probabilities and residuals.** Set $q_{t,z} = h^* E_{t,z} h$, $r_{t,z} = \mathbf{1}_{\{Z_t = z\}} - q_{t,z}$ and $\bar{q}_{t,z} = \max(q_{t,z}, \tau)$.
3. **Assemble and solve the normal equations.** Writing $\Delta = \sum_{j=1}^{2d} \beta_j T_j$ in (4.8) and setting its gradient to zero gives

$$\mathcal{K}_n = \eta \mathrm{I}_{2d} + \sum_{t,z} g_{t,z} g_{t,z}^{\mathsf{T}} / \bar{q}_{t,z}, \qquad S_n^{\mathrm{fit}} = \sum_{t,z} g_{t,z} r_{t,z} / \bar{q}_{t,z}, \qquad \mathcal{K}_n \widehat{z}_n = S_n^{\mathrm{fit}}. \tag{S2.4}$$

   The positive ridge gives $\mathcal{K}_n \succeq \eta \mathrm{I}_{2d}$, so the linear solve has a unique solution $\widehat{z}_n \in \mathbb{R}^{2d}$.
4. **Reconstruct and normalize.** Set $\widehat{\Delta}_n = \sum_{j=1}^{2d} [\widehat{z}_n]_j T_j$ and apply the normalization (4.9):

$$\widehat{h}_n^{\mathrm{eff}} = \frac{h + \widehat{\Delta}_n h}{\| h + \widehat{\Delta}_n h \|}, \qquad \widehat{P}_n^{\mathrm{eff}} = \widehat{h}_n^{\mathrm{eff}} (\widehat{h}_n^{\mathrm{eff}})^*.$$

Here $\mathcal{K}_n$ is the Hessian of the stabilized quadratic objective, a weighted Gram matrix plus the ridge term. The weighted residual vector S_n^{fit} combines the differences between observed outcome indicators and fitted probabilities along the tangent directions. When clipping is inactive, $\mathcal{K}_n - \eta \mathrm{I}_{2d}$ is the matrix of $\sum_{t \le n} \mathcal{J}_{a_t,P}$ in this basis, by (2.4). The solution $\widehat{z}_n$ contains the coefficients of the tangent correction, and normalization converts it into the reported estimator. Changing the orthonormal basis changes these coefficients but leaves $\widehat{\Delta}_n$ and $\widehat{P}_n^{\mathrm{eff}}$ unchanged.

Computational costs and reproduction details for the main figures are in Sections S9 and S10.

Part II. Supporting results and reproducibility

S3. Probability tools for adaptive experiments

The conditional observation law gives martingale differences even though the adaptive records are dependent. These lemmas support the strong laws and Gaussian limits in Sections S1.2 and S1.3.

The next lemma controls bounded martingale sums arising from adaptively selected measurements.

Lemma S3.1 (Bounded martingale sums)**.** *If X_t is adapted, $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = 0$, and $|X_t| \le C$, then $n^{-1}\sum_{t\le n} X_t \to 0$ almost surely. Moreover, $\sum_{t\le n} X_t = O_{\mathrm{a.s.}}(\{n\log(n+1)\}^{1/2})$. The same conclusions hold coordinatewise for bounded vector or matrix increments.*

Proof of Lemma S3.1. For $C > 0$, the conditional exponential bound for a bounded centered increment (Azuma, 1967) gives

$$\mathbb{P}\left(\left|\sum_{t\le n} X_t\right| \ge u\right) \le 2\exp[-u^2/(2nC^2)].$$

At $u = n\epsilon$ the bound is summable for every $\epsilon > 0$. Borel–Cantelli and a countable intersection over positive rational ϵ prove the strong law. At $u = C\{6n\log(n+1)\}^{1/2}$ the bound is $2(n+1)^{-3}$, proving the rate. The case $C = 0$ is immediate; intersect finitely many probability-one events for the vector assertion. □

The next lemma shows how finite batching and averaging preserve convergence.

Lemma S3.2 (Freezing and averaging)**.** *For (2.1), $n_-(t) \to \infty$. Thus $x_n \to x_0$ implies $x_{n_-(t)} \to x_0$. If $A_t \to A$ in a finite-dimensional space, then $n^{-1}\sum_{t\le n} A_t \to A$.*

Proof of Lemma S3.2. For each K, choose j with $N_j \ge K$. Once $t > N_j$, $n_-(t) \ge N_j$. This proves the first two assertions. To average, split the sum at a fixed index beyond which $\|A_t - A\| < \epsilon$; the finite prefix divided by n tends to zero. □

The lemma uses infinitely many finite batches. It does not describe a two-stage asymptotic experiment whose preliminary sample size stays fixed.

The next lemma establishes asymptotic normality for normalized sums of bounded martingale differences whose conditional covariance has a deterministic limit. For general martingale central limit theory, see Hall and Heyde (1980, Section 3.2); here we give a direct proof for bounded increments.

Lemma S3.3 (Direct adaptive asymptotic normality)**.** *Let X_t be bounded martingale differences in $\mathbb{R}^p$. Suppose their current conditional covariance converges almost surely to a deterministic positive semidefinite matrix:*

$$\mathbb{E}[X_t X_t^{\mathsf{T}} \mid \mathcal{F}_{t-1}] \longrightarrow V.$$

Then $n^{-1/2}\sum_{t\le n} X_t \Longrightarrow \mathcal{N}_p(0, V)$.

Proof of Lemma S3.3. Fix $u \in \mathbb{R}^p$ and a real frequency s. Put $x_t = u^{\mathsf{T}} X_t$, $v_t = \mathbb{E}[x_t^2 \mid \mathcal{F}_{t-1}]$, $v = u^{\mathsf{T}} V u$, and choose C_u with $|x_t| \le C_u$. For each sample size n let $a = s/n^{1/2}$ and $\alpha = 1 - a^2 v/2$. Conditional Taylor expansion of the complex exponential gives

$$\left|\mathbb{E}[e^{iax_t} \mid \mathcal{F}_{t-1}] - \alpha\right| \le \frac{a^2}{2}|v_t - v| + |a|^3 C_u^3 e^{|a|C_u}.$$

The linear term vanishes by conditional centering. If $f_k = \mathbb{E}\{\exp(ia\sum_{t\le k} x_t)\}$, conditioning on the past and using its exponential's unit modulus yields

$$|f_k - \alpha f_{k-1}| \le \tfrac{a^2}{2}\mathbb{E}(|v_k - v|) + |a|^3 C_u^3 e^{|a|C_u}.$$

For all large n, $0 \le \alpha \le 1$. Iteration from $f_0 = 1$ gives

$$\begin{aligned}\left|\mathbb{E}\{e^{is\sum_{t\le n} x_t/n^{1/2}}\} - \left(1 - \frac{s^2 v}{2n}\right)^n\right| \\ \le \frac{s^2}{2n}\sum_{t\le n}\mathbb{E}(|v_t - v|) + \frac{|s|^3 C_u^3 e^{|s|C_u}}{n^{1/2}}.\end{aligned} \tag{S3.1}$$

Uniform boundedness and almost-sure convergence give $\mathbb{E}(|v_t - v|) \to 0$. The right side vanishes by the averaging assertion of Lemma S3.2, and the comparison power tends to $e^{-s^2 v/2}$. Lévy's continuity theorem and Cramér–Wold prove the assertion, including directions for which $v = 0$. □

In each application we verify convergence of the current conditional covariance. Bounded increments then allow this direct characteristic-function proof. The argument is specific to these conditions and is not a substitute for a general martingale central limit theorem.

S4. Tangent and coordinate bounds

These bounds supply the geometric remainders in Sections S1.3 and S2.6. The tangent projection and retraction are defined in Appendix A.

Lemma S4.1 (Projection and normalization bounds)**.** *Let $P = hh^*$ and P_0 be pure states, with $\|h\| = 1$. For $e = P - P_0$,*

$$\|e - \Pi_P e\|_F = \|e\|_F^2/\sqrt{2}, \qquad \left\|\Pi_P - \Pi_{P_0}\right\|_{\mathrm{op}} \le 6\,\|e\|_F.$$

For $\Delta \in T_P$, the normalization retraction is

$$R_P(\Delta) = \frac{(\mathrm{I} + \Delta)P(\mathrm{I} + \Delta)}{1 + \|\Delta h\|^2}. \tag{S4.1}$$

It satisfies $\|R_P(\Delta) - P\|_F \le \|\Delta\|_F$ and $\|R_P(\Delta) - P - \Delta\|_F \le 3\|\Delta\|_F^2$. If A is Hermitian and $\mathrm{proj}_{\mathcal{P}_D}(A)$ is any nearest pure projector, then

$$\left\|\mathrm{proj}_{\mathcal{P}_D}(A) - P_0 - \Pi_{P_0}(A - P_0)\right\|_F \le 20\,\|A - P_0\|_F^2.$$

Proof of Lemma S4.1. Write $P_0 = gg^*$ and $g = \alpha h + v$, with $v \perp h$. The block-diagonal part of e has blocks $\|v\|^2$ and $-vv^*$, while $\|e\|_F^2 = 2\|v\|^2$. This proves the first identity. For the second, expand the difference of the projections applied to B as $eB + Be - 2(eBP + P_0 Be)$ and use submultiplicativity.

For $\Delta = hv^* + vh^*$, with $v \perp h$, one has $\Delta h = v$ and $\|\Delta\|_F^2 = 2\|v\|^2$. Expanding $(h+v)(h+v)^*$ proves (S4.1) and

$$\|R_P(\Delta) - P\|_F^2 = \frac{2\|v\|^2}{1+\|v\|^2} \le \|\Delta\|_F^2.$$

Subtracting $P + \Delta$ in the same formula bounds the remainder by $3\|\Delta\|_F^2$.

For the last assertion, set $E = A - P_0$ and $B = \mathrm{proj}_{\mathcal{P}_D}(A) - P_0$. The identity $\|A - P\|_F^2 = \|A\|_F^2 + 1 - 2\,\mathrm{tr}(AP)$ shows that a nearest pure projector uses a maximizing eigenvector, so $[A, \mathrm{proj}_{\mathcal{P}_D}(A)] = 0$. Since P_0 is an admissible competitor, $\|\mathrm{proj}_{\mathcal{P}_D}(A) - A\|_F \le \|P_0 - A\|_F$; the triangle inequality gives $\|B\|_F \le 2\|E\|_F$. Since $\Pi_{P_0}(X) = [P_0, [P_0, X]]$, commutation implies

$$\Pi_{P_0}(B - E) = -[P_0, [E, B]].$$

The bound $\|[X, Y]\|_F \le 2\|X\|_F\|Y\|_F$ gives $\|\Pi_{P_0}(B-E)\|_F \le 8\|E\|_F^2$. The first identity gives $\|B - \Pi_{P_0}B\|_F = \|B\|_F^2/\sqrt{2}$. Their sum is at most $20\|E\|_F^2$, including eigenvalue ties. □

Lemma S4.2 (Transfer to smooth coordinates)**.** *In the normalized amplitude chart centered at $P_0 = hh^*$, let $r(P)$ be twice the real and imaginary parts of the off-diagonal column of P in the chart's reference frame. Then*

$$r\{\Gamma_{h,b}(u)\} = \sqrt{1 - \|u\|^2/4}\,u, \qquad u(P) = r(P)/\sqrt{h^*Ph}. \tag{S4.2}$$

If $P_n \to P_0$ in probability and $\sqrt{n}\,r(P_n) \Longrightarrow \mathcal{N}_{2d}(0, 2\mathrm{I})$, then in any fixed smooth chart φ with $\theta_0 = \varphi(P_0)$,

$$\sqrt{n}\{\varphi(P_n) - \varphi(P_0)\} \Longrightarrow \mathcal{N}_{2d}(0, 2H(\theta_0)^{-1}).$$

Proof of Lemma S4.2. Multiplication of the vector in (4.2) by its adjoint proves (S4.2). Consistency gives $h^*P_nh \to 1$; hence the amplitude coordinates have the same limit by Slutsky's theorem. For the smooth transition map f to φ, write $f(u) - f(0) = Ju + \|u\|\,r(u)$, where $r(u) \to 0$. Tightness of $\sqrt{n}\,u(P_n)$ makes the remainder negligible. The covariance is $2JJ^{\mathsf{T}}$ and $H(\theta_0) = J^{-\mathsf{T}}J^{-1}$ by Proposition 2.1 and (2.6). Coordinates can be defined arbitrarily outside the chart, an event whose probability tends to zero. □

S5. The reference measurement for Algorithm 3

This construction supplies the reference POVM in Example 4.1 and main Figure 2. Fix $D \ge 2$, put $d = D - 1$, and start with the coordinate basis $e_0, \dots, e_d$. For every $r < s$, also use

$$\widetilde{\mathcal{B}}_{rs}^{\mathrm{R}} = \{(e_r + e_s)/\sqrt{2}, (e_r - e_s)/\sqrt{2}\} \cup \{e_p : p \ne r, s\},$$
$$\widetilde{\mathcal{B}}_{rs}^{\mathrm{I}} = \{(e_r + ie_s)/\sqrt{2}, (e_r - ie_s)/\sqrt{2}\} \cup \{e_p : p \ne r, s\}.$$

List these $B = 1 + 2\binom{D}{2} = D^2 - D + 1$ bases in a fixed order as $\widetilde{\mathcal{B}}_\ell = \{\widetilde{f}_{\ell,k} : k \in [D]\}$. Set

$$c_D^2 = \sum_{r=0}^{d}(r+1)^2, \qquad w = c_D^{-1}\sum_{r=0}^{d}(r+1)e_r,$$
$$u = w - e_0, \qquad R = \mathrm{I}_D - 2uu^*/(u^*u),$$
$$f_{\ell,k} = R\widetilde{f}_{\ell,k}, \qquad F_{\ell,k} = B^{-1}f_{\ell,k}f_{\ell,k}^*.$$

Proposition S5.1 (Reference basis construction)**.** *The effects $F_{\ell,k}$ satisfy (C1)–(C2) in the main paper, with*

$$\min_{\ell,k} \operatorname{tr}(P_{\mathrm{ref}} F_{\ell,k}) = 1/(2Bc_D^2).$$

After phase pairing, there are $2BD$ complete labels and the denominator floor is $\tau = 1/(16Bc_D^2)$. The measurement is implemented by one uniformly selected D-outcome PVM on each copy, with the branch and basis recorded.

Proof of Proposition S5.1. Since $u \neq 0$, $R = R^*$ is unitary and $R^* e_0 = w$. Thus each transformed basis is orthonormal and the effects sum to I_D. Before transformation, the coordinate projectors give the diagonal Hermitian units. Differences of the real and imaginary superposition projectors give

$$e_r e_s^* + e_s e_r^*, \qquad i(e_s e_r^* - e_r e_s^*),$$

respectively (Bertlmann and Krammer, 2008, Section 3.1). These span all off-diagonal Hermitian directions. Conjugation by R and multiplication by $1/B$ preserve the span, proving (C1).

For (C2), $e_0^* f_{\ell,k} = w^* \widetilde{f}_{\ell,k}$ never vanishes: the coordinates of w are positive and distinct. The smallest squared overlap is $1/(2c_D^2)$, attained by a real difference of adjacent coordinates. This proves the probability minimum. Phase pairing halves it, and the floor is one quarter of that value.

At a copy with history-determined alignment U_t, draw independently $b \sim \mathrm{Bernoulli}(1/2)$ and $\ell \sim \mathrm{Unif}([B])$, then measure the basis $\{U_t (V^b)^* f_{\ell,k} : k \in [D]\}$. The joint conditional probability of the complete label (b, ℓ, k) is

$$\frac{1}{2B} \operatorname{tr}\{P_0 U_t (V^b)^* f_{\ell,k} f_{\ell,k}^* V^b U_t^*\},$$

exactly the Born probability of the complete randomized POVM in Algorithm 3. The factor $1/(2B)$ is the setting probability; only the selected PVM's D readouts are measured. □

For $D = 4$, $B = 13$ and $c_D^2 = 30$, so $2BD = 104$ and $\tau = 1/6240$. The reference operations are fixed: each superposition basis uses a two-level balanced rotation, with a quarter phase for an imaginary basis, preceded by R^*. The adaptive and phase operations are U_t^* and V^b, as in the main algorithm.

S6. Fitted-weight cancellation

This section supplies the expansion used in the proof of Theorem 4.3. Consistency and the preliminary root-n rates are established first in Section S1.3.

Lemma S6.1 (Multinomial score covariance)**.** *For positive probabilities $q = (q_z)$, let G have rows $g_z^\top$, where g_z is the tangent derivative of q_z. With $D_q = \mathrm{diag}(q)$,*

$$G^\top D_q^{-1}(D_q - qq^\top) D_q^{-1} G = G^\top D_q^{-1} G. \tag{S6.1}$$

Proof of Lemma S6.1. Completeness gives $\sum_z g_z = 0$ and $D_q^{-1} q = \mathbf{1}$; therefore the term involving $qq^\top$ vanishes. □

This is multinomial generalized least squares (Acharya, Kypraios and Guță, 2019, Appendix A.1). When truncation is inactive, it makes the score covariance equal the information. The correction is a regularized Fisher-scoring step on the fitted tangent space; Brouste, Dutang and Noutsa Mieniedou (2021) discuss the general one-step method.

Put $P = \widehat{P}_n$, $e = P - P_0$, $e_T = \Pi_P e$, and $p_{t,z} = \operatorname{tr}(E_{t,z}P_0)$, $\varepsilon_{t,z} = \mathbf{1}_{\{Z_t=z\}} - p_{t,z}$. Write $\bar{p}_{t,z} = \max(p_{t,z}, \tau)$ and $\bar{q}_{t,z} = \max\{\operatorname{tr}(E_{t,z}P), \tau\}$. The true-state residual sum and fitted tangent normal operator are

$$S_n^0 = \sum_{t\le n,z} \bar{p}_{t,z}^{-1}\varepsilon_{t,z}E_{t,z},$$
$$\mathsf{A}_n A = \eta A + \sum_{t\le n,z} \bar{q}_{t,z}^{-1}\langle \Pi_P E_{t,z}, A\rangle_F \Pi_P E_{t,z}, \qquad A \in T_P.$$

Lemma S6.2 (Fitted-weight cancellation)**.** *Let* E *be a fixed finite informationally complete POVM and* $E_{t,z} = U_t E_z U_t^*$*, where* U_t *is determined by* $\mathcal{F}_{t-1}$ *and the outcomes follow the Born law at the pure truth* P_0*. For the fitted tangent solve* (4.8)*, fix* $\eta > 0$ *and* $0 < \tau \le 1$*. Along predetermined reporting sample sizes* $n \to \infty$*, assume that* P *is pure,* $P \to P_0$ *almost surely,* $e = O_p(n^{-1/2})$*, and all* $p_{t,z} \ge 2\tau$ *eventually almost surely. Also assume that the average true tangent Fisher information converges almost surely to the Frobenius inner product on* T_{P_0}*. Then*

$$\widehat{\Delta}_n = -e_T + \mathsf{A}_n^{-1}\Pi_P S_n^0 + O_p(n^{-1}). \tag{S6.2}$$

Moreover, extending the inverse by zero off T_P*,* $n\mathsf{A}_n^{-1}\Pi_P \to \Pi_{P_0}$ *almost surely in operator norm.*

Proof of Lemma S6.2. The true-state residual sum S_n^0 is a bounded-increment martingale. So is the operator sum

$$M_n A = -\sum_{t\le n,z} \frac{\varepsilon_{t,z}}{\bar{p}_{t,z}^2} \operatorname{tr}(E_{t,z}A)E_{t,z}.$$

Their conditional means vanish because settings and true probabilities are determined before the next outcome. Orthogonality gives $S_n^0 = O_p(\sqrt{n})$ and $M_n = O_p(\sqrt{n})$ without conditioning on the fitted state.

The assumed probability bound gives an almost surely finite t_{reg} with $p_{t,z} \ge 2\tau$ for every $t \ge t_{\mathrm{reg}}$ and every z. For all sufficiently large reporting sizes, consistency also gives $|\operatorname{tr}(E_{t,z}e)| \le \tau$ uniformly in t, z. On these rows both denominators are untruncated. The exact reciprocal identity

$$b^{-1} = a^{-1} - (b-a)/a^2 + (b-a)^2/(a^2 b)$$

applied to the fitted normal equation yields

$$\mathsf{A}_n(\widehat{\Delta}_n + e_T) = \Pi_P S_n^0 + \Pi_P M_n e_T + R_n + \eta e_T. \tag{S6.3}$$

The residual R_n contains the reciprocal quadratic remainder, the normal part $e - e_T$, and the finite initial clipped rows. Lemma S4.1 and bounded effects imply the pathwise bound

$$\|R_n\|_F \le Cn\,\|e\|_F^2 + Ct_{\mathrm{reg}}.$$

Indeed, on each untruncated row the reciprocal remainder is $O(\|e\|_F^2)$, and the normal part has the same order. The floor bounds every initial row deterministically. These bounds hold before substituting the random fitted state.

Thus $M_n e_T$, R_n and ηe_T are $O_p(1)$. Products of tight variables require no independence, and the almost surely finite t_{reg} is tight. Since $\bar{q}_{t,z} \le 1$, the frame bound gives $\mathsf{A}_n \succeq (\eta + n\lambda_{\min}(S_{\mathsf{E}}))\mathrm{I}_{T_P}$. Its inverse has norm $O(n^{-1})$, proving (S6.2).

For the final assertion, the reciprocal weights are Lipschitz and $\Pi_P \to \Pi_{P_0}$. Replacing fitted weights and projections by their true-state values changes A_n/n by $o(1)$ in operator norm. The finite clipped prefix vanishes after division by n, and the average information tends to the identity on the true tangent space. To make inversion on the moving tangent space explicit, extend A_n by zero off T_P and set $\mathsf{B}_n = \mathsf{A}_n/n + (\mathrm{I} - \Pi_P)$ on the ambient Hermitian space. Then $\mathsf{B}_n \to \mathrm{I}$ in operator norm, and

$$n\mathsf{A}_n^{-1}\Pi_P = \mathsf{B}_n^{-1}\Pi_P \longrightarrow \Pi_{P_0}.$$

The inverse on the left acts on T_P; the ambient inverse on the right is well defined by the uniform coercivity already proved. □

S7. Moment bounds for expected loss

These estimates support Corollaries 5.2 and 5.3. Their separate proofs and the expectation-transfer argument are in Section S1.4. Constants may depend on the fixed truth, dimension, reference measurement and ridge parameters; no uniformity toward the mixed-state boundary is claimed.

Lemma S7.1 (Bounded martingale moments)**.** *For finite-dimensional martingale differences with $\|X_t\| \le B$ for a fixed deterministic B, let $M_n = \sum_{t\le n} X_t$. For each fixed $r \ge 2$ and $x \ge 0$,*

$$\mathbb{P}(\|M_n\| > x) \le C\exp(-cx^2/n), \qquad \mathbb{E}\,\|M_n\|^r \le C_r n^{r/2}, \quad n \ge 1. \tag{S7.1}$$

Proof of Lemma S7.1. Apply the exponential bound in the proof of Lemma S3.1 to each coordinate, then take a finite union (Azuma, 1967).

Integrating the tail, using $\mathbb{E}Z^r = \int_0^\infty rx^{r-1}\mathbb{P}(Z > x)\,\mathrm{d}x$ for $Z \ge 0$, gives the moment bound. □

S7.1. Mixed-qubit ridge outputs

Proposition S7.2 (Mixed-qubit moment bounds)**.** *For Algorithms 1 and 2, fix an interior truth $\|\theta_0\| < 1$, $\lambda > 0$, and predetermined positive finite batches (2.1). For each fixed $r \ge 2$ and either allocation,*

$$\sup_{n\ge1}\mathbb{E}\left\|n(\lambda\mathrm{I}_3 + G_n)^{-1}\right\|^r < \infty, \qquad \sup_{n\ge1} n^{r/2}\mathbb{E}\,\|c_n - \theta_0\|^r < \infty. \tag{S7.2}$$

In particular the physical outputs satisfy

$$\sup_{n\ge1} n^2\mathbb{E}_{\rho_0}\,\|\widetilde{\rho}_n - \rho_0\|_F^4 < \infty. \tag{S7.3}$$

Proof of Proposition S7.2. Let $Q_t = \mathbb{E}(a_t a_t^\mathsf{T} \mid \mathcal{F}_{t-1})$, $D_n = G_n - \sum_{t\le n} Q_t$, and $M_n = \sum_{t\le n} a_t(Y_t - a_t^\mathsf{T}\theta_0)$. Both centered sums have bounded increments. The exact ridge identity is

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = n(\lambda \mathrm{I}_3 + G_n)^{-1}\{M_n/\sqrt{n} - \lambda\theta_0/\sqrt{n}\}. \tag{S7.4}$$

For Bures allocation, every orientation gives $Q_t = \mathrm{I}_3/3$. The event $\{\lambda_{\min}(G_n) < n/6\}$ has probability at most Ce^{-cn}. On its complement, $\|n(\lambda\mathrm{I}_3 + G_n)^{-1}\| \le 6$; on the exceptional event it is at most n/λ. Consequently this normalized inverse has uniformly bounded moments of every fixed order.

For Frobenius allocation the safeguard gives $\sum_{t\le n} Q_t \succeq n(n+1)^{-1/4}\mathrm{I}_3/3$. Applying (S7.1) to D_n and M_n in the unnormalized ridge identity shows that, for each fixed $\delta > 0$,

$$\mathbb{P}\left(\left\|\widehat{\theta}_n - \theta_0\right\| > \delta\right) + \mathbb{P}(\|c_n - \theta_0\| > \delta) \le C_\delta e^{-c_\delta\sqrt{n}} \tag{S7.5}$$

for all sufficiently large n. Indeed, outside an event of this order the Gram eigenvalue is at least $n(n+1)^{-1/4}/6$ and the residual must have norm of order $n^{3/4}$ for the displayed error to exceed δ. The truth belongs to the projection ball for all sufficiently large n, so its Euclidean projection cannot increase this error.

To improve this deterministic floor to a moment bound at order n, use a maximum over the estimates available by time n. Since the truth is interior, there are $\delta_0, \kappa > 0$ such that every allocation weight is at least κ whenever $\|c_k - \theta_0\| < \delta_0$. Put

$$C_0 = \max(3, 3/\kappa), \qquad \mathcal{R}_n = \max\left\{C_0, \max_{1\le k\le n}\sum_{\ell=1}^{3}\pi_{F,\ell}(c_k)^{-1}\right\}.$$

The safeguard bounds the sum of inverse weights at time $k \ge 1$ by $18k$; this deliberately loose polynomial bound suffices. Outside the event $\{\|c_k - \theta_0\| \ge \delta_0\}$ that sum is at most C_0. Consequently, for each fixed $r \ge 2$,

$$\sup_n \mathbb{E}\mathcal{R}_n^r \le C_0^r + \sum_{k\ge 1}(18k)^r\mathbb{P}(\|c_k - \theta_0\| \ge \delta_0) < \infty,$$

where finitely many initial terms are harmless and (S7.5) controls the remainder. Every action used by time n has weights at least $\mathcal{R}_n^{-1}$. Thus, pathwise, $G_n \succeq (n/\mathcal{R}_n - \|D_n\|)\mathrm{I}_3$. On $A_n = \{2\mathcal{R}_n\|D_n\| \le n\}$ the normalized ridge inverse has norm at most $2\mathcal{R}_n$; everywhere it has norm at most n/λ. On A_n^c,

$$n^r\mathbf{1}_{A_n^c} \le \{2\mathcal{R}_n\left\|D_n/\sqrt{n}\right\|\}^{2r}.$$

Hölder's inequality and the moment bounds for $\mathcal{R}_n$ and $D_n/\sqrt{n}$ bound the expectation on the right uniformly in n. No independence between these variables is required. This proves, for both allocations, (S7.2) for the normalized inverse. Use the $2r$ moments of this inverse and of $M_n/\sqrt{n}$ in Hölder's inequality applied to (S7.4). The resulting rth moment of $\sqrt{n}(\widehat{\theta}_n - \theta_0)$ is uniformly bounded.

For Frobenius allocation, projection is nonexpansive once its ball contains θ_0; finitely many earlier sample sizes do not affect the moment bound. For Bures allocation the safeguard changes the output only if $\|\widehat{\theta}_n\| \ge 1$. On that event the raw error is at least $1 - \|\theta_0\|$ and the reported error is $\|\theta_0\|$. Thus a fixed multiple of the raw error bounds the reported error everywhere. Finally, $\|\rho(c_n) - \rho(\theta_0)\|_F^2 = \|c_n - \theta_0\|^2/2$. This proves (S7.3) for both mixed-qubit outputs, without a restriction on the finite predetermined batch lengths.

□

S7.2. Pure-state outputs

Proposition S7.3 (Pure-state moment bounds)**.** *For Algorithm 3, fix $D \geq 2$, a rank-one reference POVM satisfying (C1)–(C2), a pure truth P_0, $\lambda, \eta > 0$, and predetermined positive finite batches (2.1), with the algorithm's denominator floor. For each fixed $r \geq 2$,*

$$\mathbb{E}\left\|\widehat{A}_n - P_0\right\|_F^r + \mathbb{E}\left\|\widehat{P}_n - P_0\right\|_F^r + \mathbb{E}\left\|\widehat{P}_n^{\mathrm{eff}} - P_0\right\|_F^r \leq C_r n^{-r/2}. \tag{S7.6}$$

In particular the corrected output satisfies (S7.3).

Proof of Proposition S7.3. Use the bounded martingale moments of Lemma S7.1.

The uniform frame bound of Section S2.4, $G_n \succeq (\lambda + n\lambda_{\min}(S_{\mathsf{E}}))\mathrm{I}$ and the ridge equation give the bound for $\widehat{A}_n$. Nearest-pure projection increases the error by at most two, giving the bound for $\widehat{P}_n$. These estimates hold for every prescribed sequence of positive finite batches.

For the correction, put $P = \widehat{P}_n$ and $e = P - P_0$. Use the complete outcome label $z \in \mathcal{Z}$ and set $p^0_{t,z} = \mathrm{tr}(E_{t,z}P_0)$, $q_{t,z} = \mathrm{tr}(E_{t,z}P)$ and $\varepsilon_{t,z} = \mathbf{1}_{\{Z_t = z\}} - p^0_{t,z}$. The function $w(u) = 1/\max(u, \tau)$ is bounded between 1 and $1/\tau$ on $[0, 1]$ and is Lipschitz with constant at most τ^{-2}. Write

$$\begin{aligned} M_n^0 &= \sum_{t,z} w(p^0_{t,z})\varepsilon_{t,z}E_{t,z}, \\ M_n(P) &= \sum_{t,z} w(q_{t,z})\varepsilon_{t,z}E_{t,z}, \\ G_n^w(P)A &= \sum_{t,z} w(q_{t,z})\,\mathrm{tr}(E_{t,z}A)E_{t,z}. \end{aligned}$$

The first sum is a bounded-increment martingale, whereas the fitted sum need not be. Deterministically, $\|M_n(P) - M_n^0\|_F \leq Cn\|e\|_F$, $\|G_n^w(P)\| \leq Cn$ and $G_n^w(P) \succeq n\lambda_{\min}(S_{\mathsf{E}})\mathrm{I}$. The fitted normal equation therefore gives

$$\left\|\widehat{\Delta}_n\right\|_F \leq C\{\left\|M_n^0\right\|_F/n + \|e\|_F\}.$$

This estimate accommodates reuse of all records without treating the fitted weights as predictable.

Lemma S4.1 bounds the distance moved by the retraction by $\|\widehat{\Delta}_n\|_F$. The preceding deterministic inequality, the preliminary moment bound and the martingale moments of M_n^0 therefore give the same rth-moment bound for $\widehat{P}_n^{\mathrm{eff}}$. No moment assumption on a random localization time or on the estimator is required. □

S8. Finite sums for the covariance example

S8.1. Measurement construction

Fix $D \geq 2$ and put $d = D - 1$. This construction and its tangent coefficients are used only to prove the covariance example (4.7) in Section S2.6; they are not assumptions of the main theorems. Repeated library vectors are labeled outcomes, not extra independent experimental observations. Decompose $\mathbb{C}^D = \mathbb{C}e_0 \oplus \mathbb{C}^d$ and put $Z = \{1, i, -1, -i\}$. Use a labeled library

$\mathcal{V}$ consisting of one copy of each coordinate vector e_j, and, for every ordered pair (j,k) and every $\eta \in Z$, the vector

$$v_{jk\eta} = \begin{cases} e_j, & j = k, \\ (e_j + \eta e_k)/2^{1/2}, & j \neq k. \end{cases}$$

Repeated vectors retain different labels. There are $L = d + 4d^2 = d(4d+1)$ labels. All vectors are unit vectors, and

$$\sum_{v \in \mathcal{V}} vv^* = (4d+1)\mathrm{I}_d = \frac{L}{d}\mathrm{I}_d. \tag{S8.1}$$

To check this identity, the coordinate labels contribute I_d, the diagonal ordered pairs contribute $4\mathrm{I}_d$, and the off-diagonal pairs contribute $4(d-1)\mathrm{I}_d$. Their cross terms cancel under the four phases. The projectors vv^* span $\mathrm{Herm}(\mathbb{C}^d)$: coordinate labels give diagonal matrix units, and differences of the $\eta = 1, -1$ and $\eta = i, -i$ projectors give the real and imaginary off-diagonal units.

Two latitudes and four phases. Set

$$\nu_- = \frac{1}{2D}, \qquad \nu_+ = \frac{3}{2D}, \qquad c = \frac{D}{8L}.$$

Both latitudes lie strictly between zero and one. For every label $z = (\nu, \zeta, v) \in \{\nu_-, \nu_+\} \times Z \times \mathcal{V}$, define

$$w_z = \nu^{1/2}\, e_0 + \zeta\{1-\nu\}^{1/2}\, v, \qquad E_z = c w_z w_z^*. \tag{S8.2}$$

Thus the construction has $8d(4d+1)$ labeled rank-one effects. No special dimension, mutually unbiased bases, or existence of symmetric informationally complete measurements is required.

The next proposition verifies informational completeness, positive reference probabilities, and Fisher symmetry for this explicit measurement.

Proposition S8.1 (Identification, positivity and Fisher symmetry)**.** *The effects in* (S8.2) *form a measurement* $\mathsf{M} = \{E_z\}$ *with real span* $\mathrm{Herm}(\mathbb{C}^D)$. *In the reference coordinates* (4.1)*, their Fisher matrix is* $\mathcal{I}_{\mathsf{M}}(0) = H(0)/2 = \mathrm{I}_{2d}/2$ *at* $P_{\mathrm{ref}} = e_0 e_0^*$. *Equivalently,* $\mathcal{J}_{\mathsf{M},P_{\mathrm{ref}}} = \mathcal{H}_{P_{\mathrm{ref}}}/2$. *Every reference outcome probability is positive, with* $\min_z \mathrm{tr}(E_z P_{\mathrm{ref}}) = 1/(16L)$.

Proof of Proposition S8.1. For completeness, first sum phases:

$$\sum_{\zeta \in Z} w_{\nu,\zeta,v} w_{\nu,\zeta,v}^* = 4\{\nu P_{\mathrm{ref}} + (1-\nu) vv^*\}.$$

Since $\nu_- + \nu_+ = 2/D$, equation (S8.1) then gives

$$\sum_z E_z = 4c\left\{\frac{2L}{D}P_{\mathrm{ref}} + \frac{2d}{D}\frac{L}{d}\mathrm{I}_d\right\} = P_{\mathrm{ref}} + \mathrm{I}_d = \mathrm{I}_D.$$

Positivity and rank one follow directly from $E_z = c w_z w_z^*$. At the reference ray, $p_z = c\nu$, which proves the stated floor.

For global identification, the phase averages at ν_- and ν_+ give $\nu_- P_{\mathrm{ref}} + (1-\nu_-)vv^*$ and $\nu_+ P_{\mathrm{ref}} + (1-\nu_+)vv^*$. Their two coefficient vectors are linearly independent because $\nu_- \neq \nu_+$.

Hence both P_{ref} and vv^* belong to the real span of the effects. The complement library already spans all complement Hermitian blocks. Differences of the $\zeta = 1, -1$ effects give the real cross block between e_0 and v; differences of the $\zeta = i, -i$ effects give the imaginary cross block. Taking $v = e_j$ supplies all missing matrix units. This proves global identification on all Hermitian matrices, which is stronger than identification restricted to pure states.

For Fisher information, use the true local chart

$$\psi(x, y) = \{1 - (\|x\|^2 + \|y\|^2)/4\}^{1/2} e_0 + \tfrac{1}{2} \sum_{j=1}^{d} (x_j + i y_j) e_j.$$

Proposition 2.1 gives quantum Fisher information I_{2d} at zero. Put

$$g(\zeta, v) = (\Re(\zeta v), \Im(\zeta v)) \in \mathbb{R}^{2d}.$$

Direct differentiation gives $\nabla p_z = c\{\nu(1-\nu)\}^{1/2} g(\zeta, v)$. Thus the contribution to the coordinate Fisher matrix defined in (2.4) is $c(1-\nu) g g^{\mathsf{T}}$. Four-phase cancellation and equation (S8.1) give

$$\sum_{\zeta, v} g g^{\mathsf{T}} = 2 \frac{L}{d} \mathrm{I}_{2d}.$$

Finally,

$$\mathcal{I}_{\mathsf{M}}(0) = c(2 - \nu_- - \nu_+) \frac{2L}{d} \mathrm{I}_{2d} = \frac{1}{2} \mathrm{I}_{2d}.$$

Each assertion is an exact finite identity. □

The two latitudes produce unequal reference probabilities, needed for the variance inflation in the main example. Fisher-symmetric pure-state measurements have precedents in Li et al. (2016); the redundant finite construction above is given explicitly rather than assumed to exist.

S8.2. Tangent coefficients

Lemma S8.2 (Tangent Gram and noise coefficients)**.** *For the effects* (S8.2)*, let S be their unweighted frame operator and let $T = T_{P_{\mathrm{ref}}}$ with Frobenius inner product. At the reference truth,*

$$\Pi_T S = \alpha_D \Pi_T, \qquad \mathrm{Cov}\{\Pi_T (E_Z - S(P_{\mathrm{ref}}))\} = \beta_D \mathrm{I}_T,$$

where

$$\alpha_D = \frac{4D - 5}{32Ld}, \qquad \beta_D = \frac{5D - 7}{256L^2 d}.$$

The same identities hold at every exactly aligned unitary rotation, with its rotated true tangent space.

Proof of Lemma S8.2. In Frobenius-orthonormal tangent coordinates, the tangent part of $E_{\nu,\zeta,v}$ is

$$e_{\nu,\zeta,v} = \sqrt{2}\, c\sqrt{\nu(1-\nu)}\, g(\zeta, v).$$

The tight-library identity gives $\sum_{\zeta,v} gg^{\mathsf{T}} = 2(L/d)\mathrm{I}_{2d}$. Summing the four phases cancels every coupling between a tangent block and a block-diagonal Hermitian matrix. Therefore

$$\alpha_D = \frac{4c^2 L}{d} \sum_{\nu\in\{\nu_-,\nu_+\}} \nu(1-\nu) = \frac{4D-5}{32Ld}.$$

The reference outcome probability is $c\nu$, and the tangent mean is zero by the same phase cancellation. Hence

$$\beta_D = \frac{4c^3 L}{d} \sum_{\nu\in\{\nu_-,\nu_+\}} \nu^2(1-\nu) = \frac{5D-7}{256L^2 d}.$$

Unitary conjugation is orthogonal on the real Hermitian space and transports the tangent projection. Both identities thus hold at every aligned frame, independently of its phase and complement basis. □

S9. Computational requirements

The mixed-qubit procedure stores its regression Gram matrix and right-hand side and performs a three-dimensional ridge solve at each update. The pure-state procedure uses single-copy measurements from the informationally complete family available at every stage. For the reference POVM constructed from B projective bases in Section 4.2, its paired implementation has $2B$ labeled basis settings and $2BD$ complete setting–outcome labels. These counts concern classical labels and available settings, not the number of simultaneously active detectors. A batch reduces how often the fitted alignment is recomputed; the selected basis or phase can still change from copy to copy. For the explicit recipe, the number of bases grows quadratically and the probability floor decreases as the inverse fifth power of dimension (Section S5). Positive probabilities in each fixed dimension therefore do not guarantee favorable conditioning as dimension grows.

Section S5 gives the fixed basis and phase operations. These specify the apparatus, not a platform-independent circuit-cost bound.

The ambient fit is a linear system on $\mathrm{Herm}(\mathbb{C}^D)$, with D^2 real coordinates. A generic dense Gram matrix uses $O(D^4)$ storage and a fresh dense factorization costs $O(D^6)$. The additional tangent solve has $2(D-1)$ unknowns, $O(D^2)$ matrix storage, and $O(D^3)$ factorization cost *after assembly*. A dense eigensolve for the leading vector costs $O(D^3)$. These separate costs should not be identified with the cost of the entire protocol.

For the weighted correction, batch setting matrices and outcome counts must be retained to evaluate the final fitted weights. Recursive linear algebra or structured frames may reduce repeated ambient work, but no such improvement is needed by our proofs. The Householder completion has a compact rank-one representation: applying it to one classical vector costs $O(D)$, while writing its dense matrix costs $O(D^2)$. These arithmetic counts do not guarantee an efficient quantum circuit for an arbitrary many-qubit alignment.

S10. Numerical reproducibility

The GitHub repository (https://github.com/HongruZhao/adaptive-quantum-tomography) supplies the simulation release `adaptive_quantum_tomography_python_20261018.zip`, which reproduces Figures 1 and 2. It contains the package `src/adaptive_tomography/`, the full configuration `configs/paper.json`, independent algorithm tests, replicate arrays, fixed apparatus matrices, source hashes and execution receipts. The recorded environment is Python 3.9.6, NumPy 1.26.4 and Matplotlib 3.9.4. After installing the package as described in its README, run from the repository root:

```
python -B -m adaptive_tomography run \
  --config configs/paper.json
python -B -m adaptive_tomography plot
```

The first command samples new records; the second reads saved arrays. The main-figure experiments use 10,000 independent Monte Carlo replicates. Exact joint setting–outcome multinomial counts implement each fixed-within-batch Born experiment. The raw fits and reported outputs use all acquired copies, including the initial batch. The current manuscript's e_3-based mixed Householder completion is used exactly. For the pure fit, ties between equal computed largest eigenvalues are resolved by projecting successive coordinate vectors onto the maximal eigenspace and taking the first nonzero projection; phase is fixed at the first maximum-modulus component.

The archived configuration records the random-number generator settings for each experiment. The main-figure settings are specified in Section 6; the saved results give the actual reporting counts. For replicate losses L_r, the plotted mean and its Monte Carlo standard error are

$$\widehat{R}_L(n) = \frac{n}{R}\sum_r L_r, \qquad \widehat{\mathrm{se}} = \frac{n}{\sqrt{R}}\left\{\frac{1}{R-1}\sum_r (L_r - \overline{L})^2\right\}^{1/2}.$$

Error bars are 1.96 times this quantity at each reporting size; reuse of Monte Carlo replicates correlates errors across sizes. The files `results/paper/samples.npz` and `results/paper/apparatus.npz` retain all replicate losses and tangent errors, fixed POVMs, labels and truths. For main Figure 2(a), `results/paper/pure_design_samples.npz` stores the two information criteria on every Algorithm 3 loss record. We average the per-record criteria after inversion; shaded bands are 1.96 times their sample standard deviations divided by $\sqrt{R}$. All configured Monte Carlo replicates enter these averages. Adaptive count records can be regenerated from the archived configuration. The code captures source hashes before sampling and checks them again at completion. Library versions can affect random draws and floating-point results; the submitted arrays are the record used here.

References

ACHARYA, A., KYPRAIOS, T. and GUŢĂ, M. (2019). A comparative study of estimation methods in quantum tomography. *J. Phys. A* **52** 234001. https://doi.org/10.1088/1751-8121/ab1958

AZUMA, K. (1967). Weighted sums of certain dependent random variables. *Tohoku Math. J. (2)* **19** 357–367. https://doi.org/10.2748/tmj/1178243286

BERTLMANN, R. A. and KRAMMER, P. (2008). Bloch vectors for qudits. *J. Phys. A* **41** 235303. https://doi.org/10.1088/1751-8113/41/23/235303

BROUSTE, A., DUTANG, C. and NOUTSA MIENIEDOU, D. (2021). OneStep: Le Cam's one-step estimation procedure. *R J.* **13** 383–394. https://doi.org/10.32614/RJ-2021-044

FACCHI, P., KULKARNI, R., MAN'KO, V. I., MARMO, G., SUDARSHAN, E. C. G. and VENTRIGLIA, F. (2010). Classical and quantum Fisher information in the geometrical formulation of quantum mechanics. *Phys. Lett. A* **374** 4801–4803. https://doi.org/10.1016/j.physleta.2010.10.005

GILL, R. D. and MASSAR, S. (2000). State estimation for large ensembles. *Phys. Rev. A* **61** 042312. https://doi.org/10.1103/PhysRevA.61.042312

HALL, P. and HEYDE, C. C. (1980). *Martingale Limit Theory and Its Application*. Academic Press, New York.

LAPACK (2026). ZLARFG: Complex elementary reflector. Netlib documentation. Accessed October 18, 2026.

LI, N., FERRIE, C., GROSS, J. A., KALEV, A. and CAVES, C. M. (2016). Fisher-symmetric informationally complete measurements for pure states. *Phys. Rev. Lett.* **116** 180402. https://doi.org/10.1103/PhysRevLett.116.180402

MISZCZAK, J. A., PUCHAŁA, Z., HORODECKI, P., UHLMANN, A. and ŻYCZKOWSKI, K. (2009). Sub- and super-fidelity as bounds for quantum fidelity. *Quantum Inf. Comput.* **9** 103–130.

SCOTT, A. J. (2006). Tight informationally complete quantum measurements. *J. Phys. A: Math. Gen.* **39** 13507–13530. https://doi.org/10.1088/0305-4470/39/43/009

ZHU, H. and HAYASHI, M. (2018). Universally Fisher-symmetric informationally complete measurements. *Phys. Rev. Lett.* **120** 030404. https://doi.org/10.1103/PhysRevLett.120.030404